



MATHEMATICS (EXTENSION 1)

Preliminary Course Assessment Task 3

June 15, 2012

General instructions

- Working time – 50 minutes.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Diagrams shown are not necessarily to scale.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper for submission to examination supervisors.

SECTION I

- Mark your answers to this section on page 3.

SECTION II

- Answer this section in the writing booklets provided.
- Commence each new question on a new page, writing on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

NAME: # BOOKLETS USED:

Class (please ✓)

☐ 11M3A – Mr Lin/Mr Rezcallah

☐ 11M3D – Mr Lucas

☐ 11M3B – Mr Fletcher

☐ 11M3E – Mr Lam

☐ 11M3C – Ms Ziaziaris

☐ 11M3F – Mr Berry

Marker's use only.

QUESTION(S)	1-4	5	6	7	8	Total	%
MARKS	$\frac{4}{4}$	$\frac{10}{10}$	$\frac{10}{10}$	$\frac{12}{12}$	$\frac{12}{12}$	$\frac{48}{48}$	

Section I: Objective response

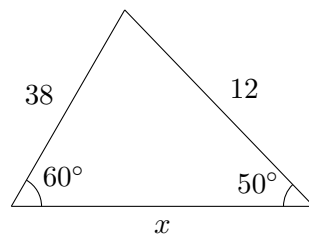
Mark your answer on page 3.

Marks

1. What is the size of each exterior angle in a regular pentagon? 1

(A) 72° (B) 108° (C) 252° (D) 540°

2. Which of the following expressions could be used to calculate the value of x ? 1



(A) $x^2 = 38^2 + 12^2 - 2 \times 12 \times 38 \times \cos 50^\circ$

(B) $x^2 = 38^2 + 12^2 - 2 \times 12 \times 38 \times \cos 60^\circ$

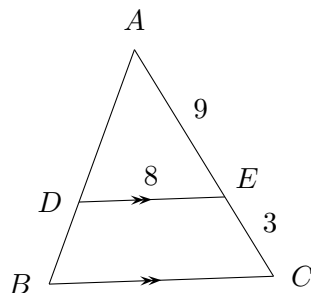
(C) $x^2 = 38^2 + 12^2 - 2 \times 12 \times 38 \times \cos 70^\circ$

(D) $\frac{x}{\sin 60^\circ} = \frac{38}{\sin 50^\circ}$

3. What is the gradient of any line perpendicular to the interval joining $(-1, 2)$ and $(4, 4)$? 1

(A) $-\frac{2}{5}$ (B) $-\frac{5}{2}$ (C) $\frac{2}{5}$ (D) $\frac{5}{2}$

4. The diagram shows $\triangle ABC$ in which $DE \parallel BC$. $AE = 9$, $EC = 3$ and $DE = 8$. 1



What is the length of BC ?

(A) 2 (B) $2\frac{2}{3}$ (C) 10 (D) $10\frac{2}{3}$

Answer page for Section I

Mark answers to Section I by fully blackening the correct circle, e.g. “●”

- | | | | | | |
|----------|---|-------------------------|-------------------------|-------------------------|-------------------------|
| 1 | — | ☐ A | ☐ B | ☐ C | ☐ D |
| 2 | — | ☐ A | ☐ B | ☐ C | ☐ D |
| 3 | — | ☐ A | ☐ B | ☐ C | ☐ D |
| 4 | — | ☐ A | ☐ B | ☐ C | ☐ D |

Examination continues overleaf...

Section II: Short answer

Question 5 (10 Marks) Commence a NEW page. **Marks**

The vertices of a triangle are $A(-1, 0)$, $B(2, 6)$ and $C(x, y)$. Given that C lies on the x axis to the right of A such that $\angle BAC = \angle BCA = \theta$,

- (a) Draw a diagram representing the above information. **2**
- (b) Find the gradient of AB and hence, find θ correct to the nearest degree. **2**
- (c) Show that the equation of the line AB is $y = 2x + 2$. **1**
- (d) Explain why BC has a gradient of -2 . **1**
- (e) If D is the point $(-2, 5)$ on this diagram, find the perpendicular distance of D from the line AB . **2**
- (f) Hence, calculate the area of the quadrilateral $ADBC$. **2**

Question 6 (10 Marks) Commence a NEW page. **Marks**

- (a) Find the coordinates of the point P that divides the interval AB externally in the ratio $1 : 4$, where A is $(3, 1)$ and B is $(-1, -5)$. **3**
- (b) Sketch the region where the following inequalities hold simultaneously: **3**

$$\begin{cases} y > x^2 \\ y \leq \sqrt{4 - x^2} \\ x \geq 0 \end{cases}$$

Show all intercepts with the axes but do *not* find the intersection points of the curves.

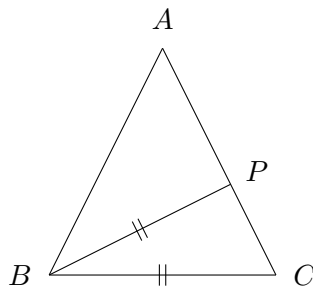
- (c) Find the equation of the line passing through the intersection of $2x - y + 1 = 0$ and $x + 3y - 1 = 0$, and perpendicular to the line $3x - 2y + 4 = 0$. **4**

Question 7 (12 Marks)

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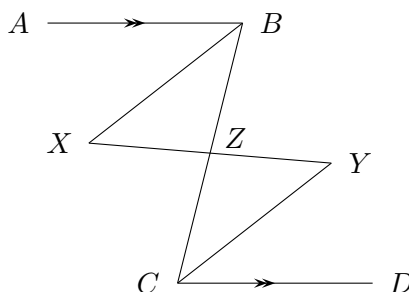
Marks

- (a) $\triangle ABC$ is isosceles with $AB = AC$, P lies on AC such that $\angle ABP = 3\angle PBC$ and $BP = BC$. **3**



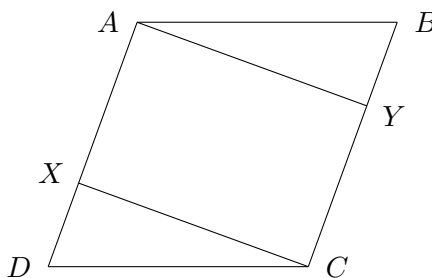
By letting $\angle CBP = x$ or otherwise, find the value of $\angle CBP$.

- (b) In this diagram, $BX = CY$, AB is parallel to CD . XB bisects $\angle ABC$ and YC bisects $\angle BCD$. **3**



Show that Z is the midpoint of BC .

- (c) $ABCD$ is a rhombus. X and Y lie on AD and BC such that $XC \perp AD$ and $AY \perp BC$.



- i. Prove that $\triangle ABY \equiv \triangle DXC$. **2**
- ii. If the side length of the rhombus is 10 cm and $\angle DAB = 120^\circ$, find the length of the diagonal AC . **2**
- iii. Find the area of the rectangle $AYCX$. **2**

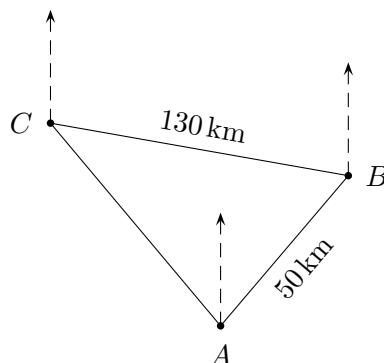
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Question 8 (12 Marks)

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Marks

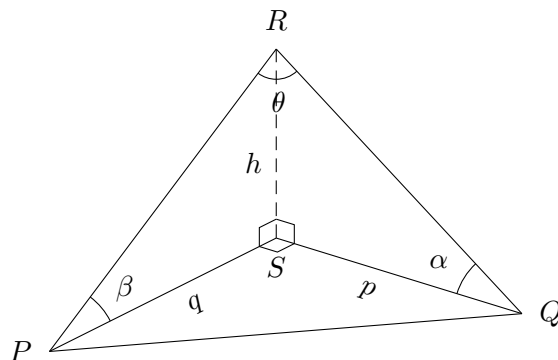
- (a) A ship sails 50 km from port A to port B on a bearing of 063° . It then sails 130 km from port B to port C on a bearing of 296° .



- i. Show, with full working, that $\angle ABC = 53^\circ$. 1
 - ii. Find AC to the nearest kilometre. 2
- (b) Sketch the graph of $y = 1 - \sin 2x$ for $-180^\circ \leq x \leq 180^\circ$. 2
- (c) A triangular pyramid has the following properties:

- | | | |
|------------|-------------------------|-------------------------|
| • $QS = p$ | • $RS = h$ | • $\angle RQS = \alpha$ |
| • $PS = q$ | • $\angle PRQ = \theta$ | • $\angle RPS = \beta$ |

Also, RS is perpendicular to the plane containing $\triangle PQS$ which is right angled at S .



- i. Show that $p = h \cot \alpha$ and obtain a similar expression for q . 2
- ii. Show that $\cos \theta = \frac{h^2}{\sqrt{(p^2 + h^2)(q^2 + h^2)}}$. 3
- iii. Hence, show that $\cos \theta = \sin \alpha \sin \beta$. 2

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Suggested Solutions

Section I

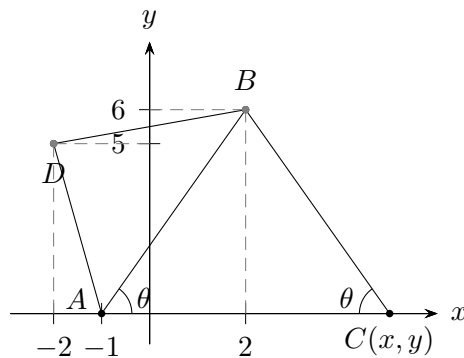
1. (A) 2. (C) 3. (B) 4. (D)

Section II

Question 5 (Lam)

- (a) (2 marks)

✓ [-1] for each missing point.



D should be filled in upon reaching part (e), but do not penalise if missing.

- (b) (2 marks)

✓ [1] for gradient.

✓ [1] for θ .

$$m_{AB} = \frac{6-0}{2+1} = \frac{6}{3} = 2$$

$$\therefore \tan \theta = 2$$

$$\theta \approx 63^\circ$$

- (c) (1 mark)

Equation of AB :

$$\frac{y-0}{x+1} = 2$$

$$y = 2(x+1) = 2x+2$$

Alternatively, use $B(2,6)$ as point in point-gradient formula.

- (d) (1 mark)

As $\angle BCA = 63^\circ$, its supplement is 117° .

$$m_{BC} = \tan 117^\circ = -\tan 63^\circ = -2$$

Alternatively, $\triangle ABC$ is isosceles, hence $C(5,0)$ and use gradient formula to demonstrate $m_{BC} = -2$.

- (e) (2 marks)

✓ [-1] for each error.

$$2x - y + 2 = 0 \rightarrow D(-2, 5)$$

$$d_{\perp} = \frac{|2(-2) + (-1)(5) + 2|}{\sqrt{2^2 + (-1)^2}}$$

$$= \frac{|-7|}{\sqrt{5}} = \frac{7}{\sqrt{5}}$$

- (f) (2 marks)

✓ [-1] for each error.

$$A_{ABCD} = A_{\triangle ABD} + A_{\triangle ABC}$$

$$A_{\triangle ABC} = \frac{1}{2} \times 6 \times 6 = 18$$

Find length AB :

$$AB = \sqrt{(2-(-1))^2 + (6-0)^2}$$

$$= \sqrt{3^2 + 6^2} = \sqrt{45} = 3\sqrt{5}$$

$$A_{\triangle ABD} = \frac{1}{2} \times d_{\perp} \times AB$$

$$= \frac{1}{2} \times \frac{7}{\sqrt{5}} \times 3\sqrt{5}$$

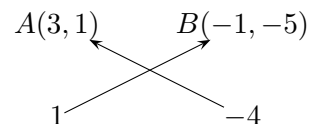
$$= \frac{21}{2}$$

$$\therefore A_{ABCD} = \frac{21}{2} + 18 = \frac{57}{2}$$

Question 6 (Ziaziaris)

- (a) (3 marks)

✓ [-1] for each error.



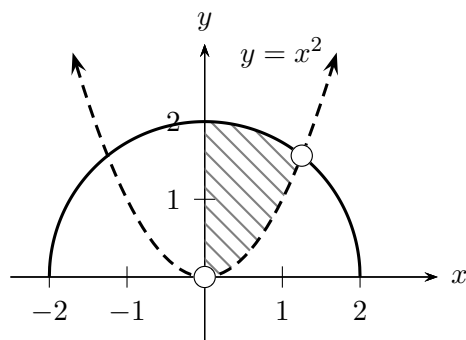
$$x_B = \frac{-4(3) + (-1)}{1-4} = \frac{-13}{-3} = \frac{13}{3}$$

$$y_B = \frac{-4(1) + (-5)}{1-4} = \frac{-9}{-3} = 3$$

$$\therefore B\left(\frac{13}{3}, 3\right)$$

(b) (3 marks)

✓ [-1] for each error.



(c) (4 marks)

Required line will be in the form

$$2x - y + 1 + k(x + 3y - 1) = 0$$

$$(2 + k)x + (-1 + 3k)y + (1 - k) = 0$$

Gradient:

$$m = \frac{2 + k}{1 - 3k} = -\frac{2}{3}$$

$$\therefore 3(2 + k) = -2(1 - 3k)$$

$$6 + 3k = -2 + 6k$$

$$3k = 8$$

$$\therefore k = \frac{8}{3}$$

Hence the line is

$$\underbrace{2x - y + 1}_{\times 3} + \underbrace{\frac{8}{3}(x + 3y - 1)}_{\times 3} = \underbrace{0}_{\times 3}$$

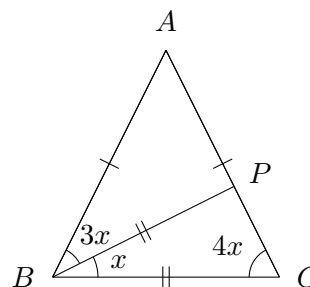
$$6x - 3y + 3 + 8x + 24y - 8 = 0$$

$$14x + 21y - 5 = 0$$

Alternatively, find the point of intersection $(-\frac{2}{7}, \frac{3}{7})$ and use point gradient formula.**Question 7** (Lin)

(a) (3 marks)

- ✓ [1] for $\angle ACB = 4x$, with reason.
- ✓ [1] for $\angle CPB = 4x$, with reason.
- ✓ [1] for final answer.



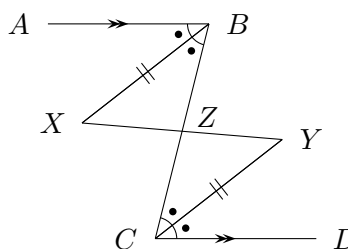
- As $\angle CBP = x$, then $\angle PBA = 3x$ (given).
- $\therefore \angle BCP = 4x$ (base $\angle$ of isos $\triangle ABC$)
- $\therefore \angle CPB = 4x$ (base $\angle$ of isos $\triangle CPB$)
- Hence $4x + 4x + x = 180^\circ$ ($\angle$ sum of $\triangle PBC$)

$$9x = 180^\circ$$

$$x = 20^\circ$$

(b) (3 marks)

✓ [-1] for each missing line or reason.

In $\triangle XBZ$ and $\triangle YCZ$,

- $XB = YC$ (given)
- $\angle XZB = \angle YZC$ (vertically opp)
- $\angle ABZ = \angle DCB$
(alternate $\angle$, $AB \parallel CD$)

 $\therefore \triangle XBZ \equiv \triangle YCZ$ (AAS)Hence $BZ = CZ$ (corresponding sides of congruent $\triangle$) and Z is the midpoint of BC .

(c) i. (2 marks)

In $\triangle ABY$ and $\triangle DXC$:

- $AB = CD$
(opp. sides of rhombus)
 - $\angle ABY = \angle CDX$
(opp. $\angle$ of rhombus)
 - $\angle AYB = 90^\circ = \angle CXD$ (given)
- $$\therefore \triangle ABY \equiv \triangle DXC$$
- (AAS)

ii. (2 marks)

- $\angle ABC = 60^\circ$ (cointerior $\angle$, $AD \parallel BC$)
- $\angle BAC = 60^\circ$ (diagonal bisects $\angle DAB$)

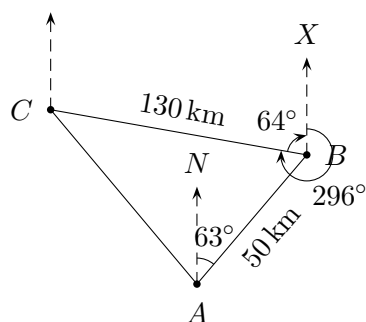
Hence $\triangle ABC$ is equilateral and $AC = 10$ cm.

iii. (2 marks)

- $AX = 5$ cm ($\triangle ADC$ equilateral, $CX \perp AD$)
- $XC = \sqrt{10^2 + 5^2} = 5\sqrt{3}$
- $A_{AYCX} = 5 \times 5\sqrt{3} = 25\sqrt{3}$

Question 8 (Berry)

(a) i. (1 mark)



- $\angle XBC = 360^\circ - 296^\circ = 64^\circ$
- $\angle NAB = 63^\circ$.
- $\therefore \angle ABC + 63^\circ + 64^\circ = 180^\circ$ (cointerior $\angle$, $NA \parallel XB$).

$$\angle ABC = 53^\circ$$

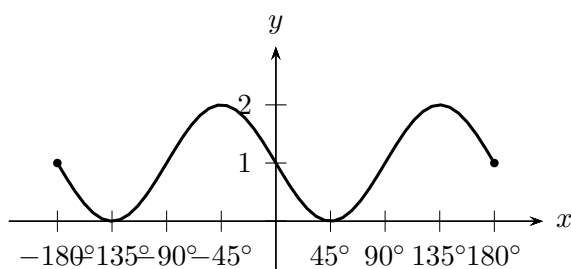
ii. (2 marks)

Using the cosine rule,

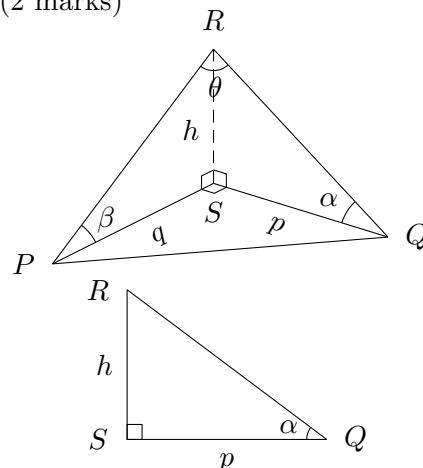
$$AC^2 = 130^2 + 50^2 - 2(130)(50)(\cos 53^\circ)$$

$$\therefore AC = 108 \text{ km}$$

(b) (2 marks)



(c) i. (2 marks)

In $\triangle RSQ$,

$$\frac{h}{p} = \tan \alpha$$

$$\therefore p = h \cot \alpha$$

Similarly in $\triangle RSP$,

$$q = h \cot \beta$$

ii. (3 marks)

- ✓ [1] for PQ^2 .
- ✓ [1] for substituting into cosine rule.
- ✓ [1] for final answer.

In $\triangle PSR$,

$$PR^2 = h^2 + q^2$$

In $\triangle QSR$,

$$QR^2 = h^2 + p^2$$

In $\triangle PRQ$,

$$PQ^2 = p^2 + q^2$$

Using the cosine rule in $\triangle PQR$,

$$PQ^2 = PR^2 + QR^2 - 2(PR)(QR) \cos \theta$$

$$\cancel{p^2 + q^2} = (h^2 + \cancel{q^2}) + (h^2 + \cancel{p^2}) - 2\sqrt{(h^2 + q^2)(h^2 + p^2)} \cos \theta$$

$$\therefore \cancel{2}h^2 = \cancel{2}\sqrt{(h^2 + q^2)(h^2 + p^2)} \cos \theta$$

$$\therefore \cos \theta = \frac{h^2}{\sqrt{(h^2 + q^2)(h^2 + p^2)}}$$

iii. (2 marks)

$$\sin \alpha = \frac{h}{\sqrt{h^2 + p^2}} \quad \sin \beta = \frac{h}{\sqrt{h^2 + q^2}}$$

$$\therefore \sin \alpha \sin \beta = \frac{h^2}{\sqrt{(p^2 + q^2)(q^2 + h^2)}} = \cos \theta$$